SECTION-I

Single Correct Answer Type 9 Q. [3 M (–1)]

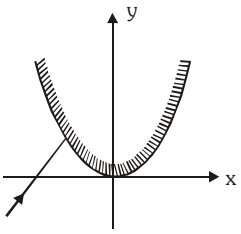
1. The apparent depth of water in cylindrical water tank of diameter 2R cm is reducing at the rate of x cm/minute when water is being drained out at a constant rate. The amount of water drained in c.c. per minute is (n_1 = refractive index of air, n_2 = refractive index of water) :-

(A) $x \pi R^2 n_1/n_2$ (B) $x \pi R^2 n_2/n_1$ (C) $2 \pi R n_1/n_2$ (D) $\pi R^2 x$

2. A light ray incident along vector $2\hat{i}+4\hat{j}+\sqrt{5}\hat{k}$ strikes on the x-z plane from medium-I of refractive index 2 and enters intomedium-II of refractive index μ_2. The value of μ_2 for which the ray is just totally reflected from the boundary, is

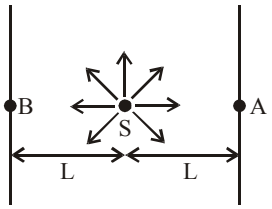
(A) $\frac{5}{4}$ (B) $\frac{6}{5}$ (C) $\frac{3\sqrt{3}}{5}$ (D) 8

3. You are given a parabolic mirror whose inner surface is silvered. The equation of the curve formed by its intersection with x–y plane is given by $y= x^2/4$. A ray travelling in x–y plane along line $y= x+3$ hits the mirror in second quadrant and gets reflected. The unit vector in the direction of reflected ray is :-



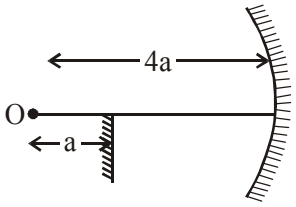
(A) $\frac{1}{\sqrt{2}}\left(-\hat{i}-\hat{j}\right)$ (B) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$ (C) $\frac{1}{\sqrt{2}}\left(-\hat{i}+\hat{j}\right)$ (D) None of these

4. The intensity of light from a source decreases with the distance x from the source, as $\frac{1}{x^2}$. In the following picture, the intensity of light from the source S at point A on a curtain is 8.1 U, where U is some unit. We then add a big mirror parallel to the curtain at point B. The distances AS and SB are equal, and the points A, S and B are in line, perpendicular to the curtain and the mirror. What is the intensity in U of light reaching point A now ?



(A) 9 (B) 15 (C) 12 (D) 18

5. The diagram shows a mirror system. If after two reflections the image and the object O coincide with each other, then the value of radius of curvature of the concave mirror is : (Take a = 3m)



(A) 4 m

(B) 6 m

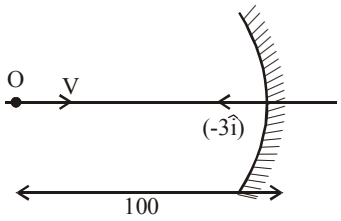
(C) 8 m

(D) 10 m

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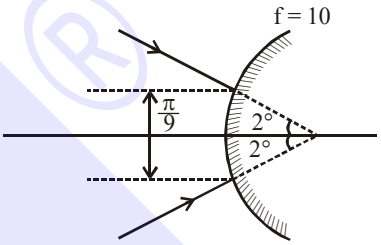
6. Radius of curvature of a concave mirror varies as $R = \left|\frac{6}{t^2 - 5t + 6}\right| cm$.

Initially the object was at a distance 100 cm from the pole as shown. At t = 0 the object started moving with velocity $v = 2t\hat{i}\,(cm/s)$. Find the speed of image at time t = 3 sec. Mirror is moving with constant velocity $\left(-3\hat{i}\right)$.



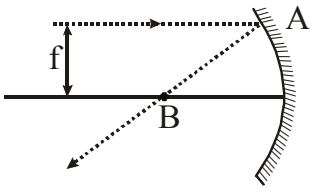
(A) 12 cm/s (B) 20 cm/s (C) 5 cm/s (D) 24 cm/s

7. A converging beam of light having angle of convergence 4° is incident upon a convex mirror as shown. Find the angle of convergence after reflection. Focal length of mirror is 10 cm.



(A) 0.5°
(B) 1°
(C) 1.5°
(D) 2°

8. A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B to the distance of the focus from the centre of curvature is (AB is the reflected ray) :-



(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{2}{3}$ (D) $\frac{1}{2}$

9. A person has D cm wide face and his two eyes are separated by d cm. The minimum width of a mirror required for the person to view his complete face is

(A) $\frac{D+d}{2}$ (B) $\frac{D-d}{4}$ (C) $\frac{D+d}{4}$ (D) $\frac{D-d}{2}$

Multiple Correct Answer Type 8 Q. [4 M (–1)]

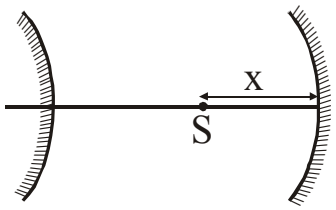
10. Two mirror's are inclined at an angle of 30° one of the ray is incident parallel to the one of the mirror. Then select the correct statement(s)-
(A) Ray will undergo five reflections from the system
(B) Total deviation from the system will 120°
(C) Ray will retrace it path finally
(D) Total deviation suffered by the ray will be 90°

11. A ray of light travelling in the direction $\frac{1}{5}(3\hat{i} + 4\hat{j})$ is incident on a plane mirror. After reflection, it travel along the direction $\frac{1}{5}(3\hat{i} - 4\hat{j})$. Then choose the correct statement(s).

(A) Mirror is kept in x-y plane (B) Normal of the mirror is along $-\hat{j}$
(C) Angle of incidence is 37° (D) Mirror is kept in x-z plane

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12. A convex mirror and a concave mirror are placed on the same optic axis, separated by distance L = 20cm. The radius of curvature of each mirror has a magnitude of 12 cm. A light source is located at a distance x from the concave mirror as shown in figure.



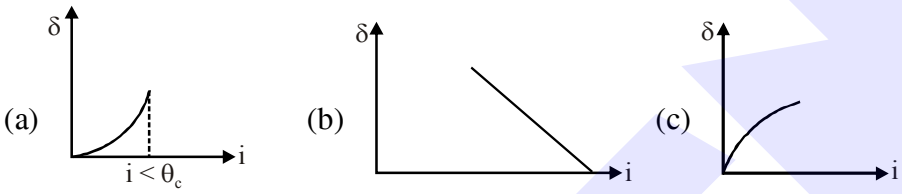
(A) The rays from the source return to the source after reflecting first from the convex mirror and then from the concave mirror for x = 8 cm.

(B) The rays from the source return to the source after reflecting first from the convex mirror and then from the concave mirror for x = 16 cm.

(C) If the rays from the source return to the source after first reflecting from the concave mirror and then from the convex mirror for x = 8 cm.

(D) If the rays from the source return to the source after first reflecting from the concave mirror and then from the convex mirror for x = 16 cm.

13. Following are graphs of angle of deivation versus angle of incidence.



Based on the above graphs mark the correct options.

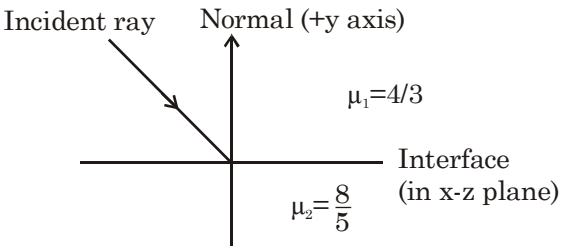
(A) Graph-a may be a part of the graph for ray of light that travels from denser to rarer medium.

(B) Graph-b may be for ray of light that is totally internally reflected from a denser to rarer medium boundary.

(C) Graph-c may be a part of the graph for ray of light that travels from rarer to denser medium.

(D) Graph-b may be a part of the graph for ray of light that is reflected from a plane mirror.

14. A light ray incident in a medium of refractive index $\frac{4}{3}$ is shown in diagram. Interface seperating the two mediums lies in xz plane. Vector along incident ray is $3\hat{i} - 4\hat{j}$.



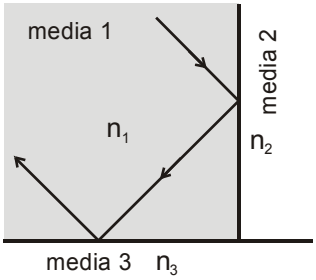
(A) Angle of incidence is 37°.

(B) The angle of refraction is 60º.

(C) The angle of refraction is 30°

(D) If angle of incidence of the incident ray is greater than the critical angle then total internal reflection takes place.

15. In the diagram shown, light is incident on the interface between media 1 (refractive index n_1) and 2 (refractive index n_2) at angle slightly greater than the critical angle, and is totally reflected. The light is then also totally reflected at the interface between media 1 and 3 (refractive index n_3), after which it travels in a direction opposite to its initial direction. The media must have a refractive indices such that



(A) $n_1 < n_2 < n_3$

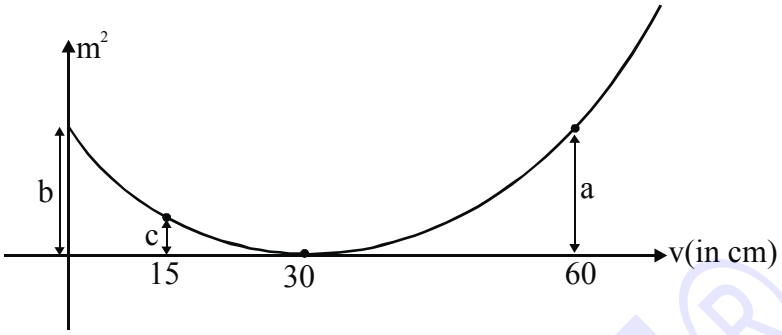
(B) $n_1^2 - n_3^2 > n_2^2$

(C) $n_1^2 - n_2^2 < n_3^2$

(D) $n_1^2 + n_2^2 > n_3^2$

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16. For a concave mirror, graph of square of magnification and image distance from pole is given for real object. Choose the correct statement(s).



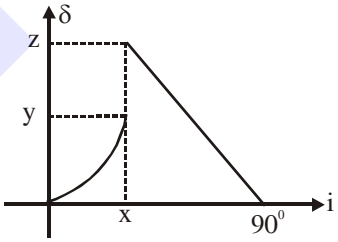
(A) This graph is hyperbola.

(B) length a must be equal to length b.

(C) radius of concave mirror is 60 cm

(D) the length c will be half of length a.

17. Light is incident from glass to air. The variation of the angle of deviation d with the angle of incident i for $0 < i < 90°$ is shown. The values of x,y and z, in terms of critical angle C, shown in the figure are



(A) x = C

(B) y = 90º–C

(C) z = 180º– 2C

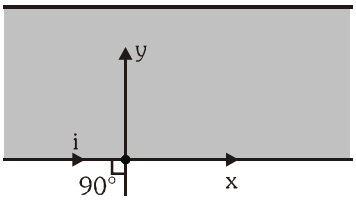
(D) z = 180° – C

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

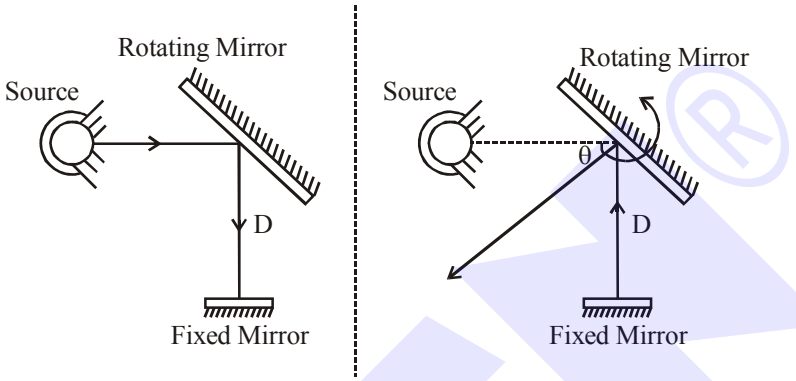
18. A ray of light is falling on a glass sphere of $\mu = \sqrt{3}$ such that the incident ray and the emergent ray, when produced, intersect at a point on the surface of the sphere. The angle of incidence in degrees is found to be $\frac{2\pi}{k}$. Find the value of k.

19. In direction of y, medium become denser as $\mu = 1 + y^2$ above x–axis. A light ray enters at an angle almost at 90° at origin. At what value of y, the slope of path of light is $\sqrt{3}$.

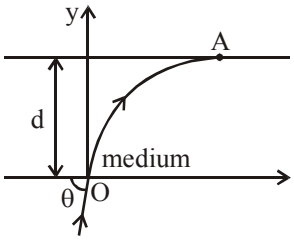


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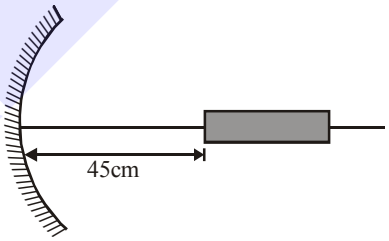
20. In the mid-19th century, French physicist Foucault made the most accurate measurement of the speed of light using a laboratory-sized apparatus consisting of rotating and fixed mirrors. Consider a source of light emitting a beam which bounces off a rotating mirror and then reflects off a small fixed mirror. By the time it reaches back to the rotating mirror, the mirror has rotated by a small angle. This rotation will deflect the beam through a small angle 30 minute of an arc from its original path. If the mirror rotates with a frequency f ((5/3) x 10^4 Hz) and the speed of light is given as 3×10^8 m/sec , the value of D (where D is distance between rotating mirror and fixed mirror) is $\frac{5}{4}\alpha$ m then the value of α is



21. A long rectangular slab of transparent medium of thickness d is placed on table with length parallel to x-axis and width parallel to the y-axis. A ray of light travelling along y-axis enters at origin such that angle of incidence θ ($\theta \to 90°$). The refractive index μ of the medium varies as $\mu = \sqrt{1+e^{x/d}}$. The refractive index of the air is 1. If the x-coordinate of the point A, where the ray intersects the upper surface of the slab-air boundary is (kd) $\ell n2$, then find the value of k.



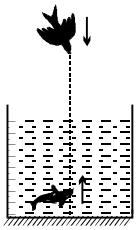
22. An object of length 30 cm is placed on principal axis of a concave mirror of focal length 30 cm. Its one end at a distance of 45 cm as shown. If length of image is 10x (in cm) find the value of x.



SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0))]**

23. A bird in air is diving vertically over a tank with speed 6 cm/s. Base of the tank is silvered. A fish in the tank is rising upward along the same line with speed 4 cm/s. [Take: $\mu_{water} = 4/3$]



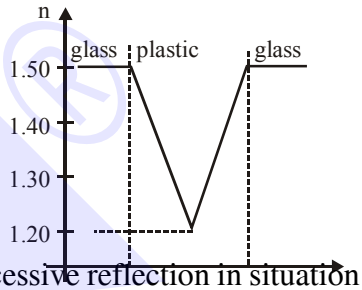
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	Column I	Column II (in cm/s)	
(A)	Speed of the image of fish as seen by the bird directly	(P)	12
(B)	Speed of the image of fish formed after reflection from the mirror as seen by the bird	(Q)	4
(C)	Speed of image of bird relative to the fish looking upwards	(R)	9
(D)	Speed of image of bird relative to the fish looking downwards in the mirror	(S)	3

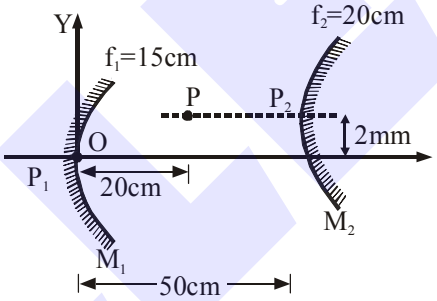
Subjective Type

4 Q. [4 M (0)]

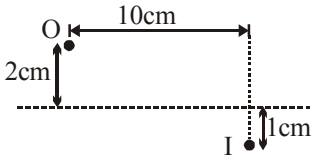
24. A thin plate of transparent plastic is embedded in a thick slab of glass. The index of refraction of the glass is n = 1.50 ; the index of refraction of the plate changes as shown in the diagram. A beam of light passes through glass and strikes the surface of the plastic plate. The maximum angle of incidence θ_{max} enables the beam to pass through the plate. Find θ_{max}.



25. Find the co-ordinates of image of point object P formed after two successive reflection in situation as shown in figure considering first reflection at concave mirror and then at convex.



26. The principal axis of a spherical mirror is shown by dotted line. O is the point object whose real image is I. Find the distance of the pole and centre of curvature of the mirror from object measured along principal axis by drawing ray diagram.



27. The x-y plane is the boundary between two transparent media. Medium-1 with z > 0 has refractive index $\sqrt{2}$ and medium – 2 with z < 0 has a refractive index $\sqrt{3}$. A ray of light in medium –1 given by the vector $A = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. Find the unit vector in the direction of refracted ray in medium -2.

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GR # GEOMETRICAL OPTICS-I **ANSWER KEY**

SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (B) **2. Ans. (B)** **3. Ans. (A)** **4. Ans. (A)** **5. Ans. (C)** **6. Ans. (A)**
7. Ans. (D) **8. Ans. (A)** **9. Ans. (D)**

Multiple Correct Answer Type **8 Q. [4 M (–1)]**

10. Ans. (A,C) **11. Ans. (B,C,D)** **12. Ans. (A,C)** **13. Ans. (A,B,D)** **14. Ans. (A,C)**
15. Ans. (B,D) **16. Ans. (B,C)** **17. Ans. (A, B, C)**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

18. Ans. 6 **19. Ans. 1** **20. Ans. 5** **21. Ans. 2** **22. Ans. 4**

SECTION-IV

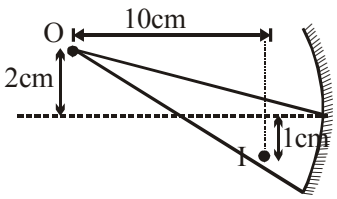
Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

23. Ans. (A) – R; (B) – S; (C) – P; (D) – Q

Subjective Type **4 Q. [4 M (0)]**

24. Ans. $\theta \approx 53^\circ$ **25. Ans. (30 cm , – 14 mm)**

26. Ans. 20 cm from object, 20/3 cm from object



27. Ans. $\vec{r} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{2\sqrt{2}}{5}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}\,(\text{angle of incidence} = 60^0; r = 45^0)$

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